Question 4

(a) Appraise the use of SDL for geometry rendering. SDL has primitive functions for drawing point, lines and rectangles. Consider the following `drawShape` function which is responsible for producing the output in Figure Q4(a).

```
void drawShape() {
    SDL_Rect fillRect = { 10, 10, 80, 80 };
    SDL_RenderFillRect(gRenderer, &fillRect);

    SDL_Rect outlineRect = { 110, 10, 80, 80 };
    SDL_RenderDrawRect(gRenderer, &outlineRect);

    SDL_SetRenderDrawColor(gRenderer, 0x00, 0x00, 0x00, 0xFF);
    SDL_RenderDrawLine(gRenderer, 250, 10, 210, 90);
    SDL_RenderDrawLine(gRenderer, 250, 10, 290, 90);
    SDL_RenderDrawLine(gRenderer, 210, 90, 290, 90);
}
```

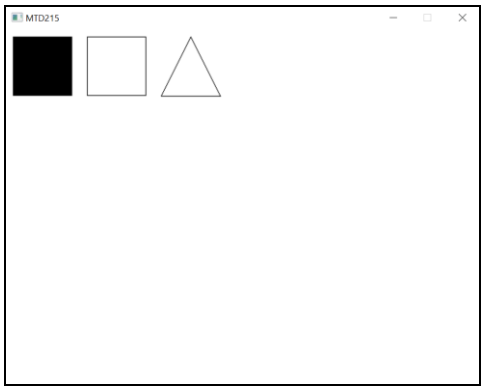


Figure Q4(a)

Re-write the `drawShape` function to produce each of following output:

(i) Sixteen Squares

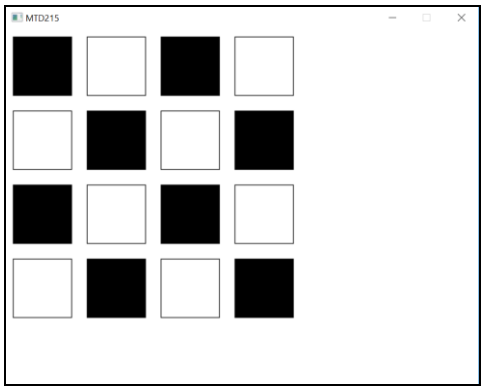


Figure Q4(a)(i)

(10 marks)

(ii) Ten Triangles

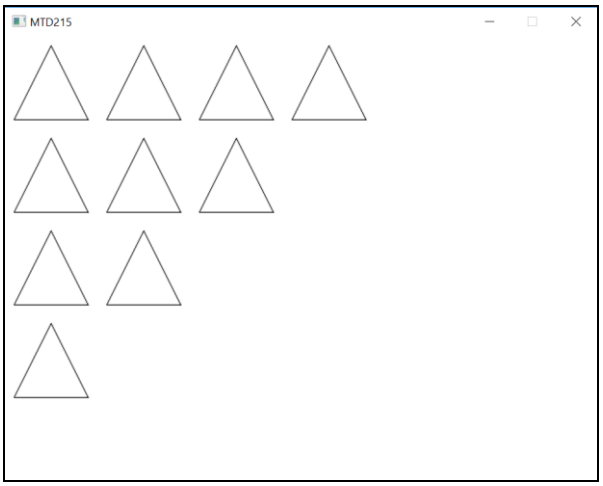


Figure Q4(a)(ii)

(8 marks)

(b) The `SDL_ttf` library provides a number of functions that allow a C++ program to initialise, load, open and close a true type font and render text onto a `SDL_Surface` object.

Identify and state the functions in the `SDL_ttf` library that are responsible for the above tasks.

(7 marks)

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